

No. of Functional Features Included in the Solution

Date	15/07/2024
Team ID	SWTID-2026-3095
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S. No	Feature Name	Feature Description	Module / Component	Status	Marks Contribution
1	Applicant Data Input Form	Web form collecting demographic, income, employment, family, housing, and contact details from the applicant.	Frontend (predict.html)	Done	High
2	Categorical Encoding & Occupation Remapping	Encodes categorical fields to match training-time format; remaps rare occupation values to "Other" so unseen categories don't break inference.	Backend (app.py)	Done	High
3	Feature Engineering Recreation	Recreates engineered features at inference time - log income, income per person, contact score, and employment flag.	Backend (app.py)	Done	Medium
4	Column Alignment (Reindexing)	Aligns the encoded input row to the exact column order used at training time via model_columns.pkl, filling missing dummy columns with 0.	Backend (app.py)	Done	High
5	XGBoost Risk Prediction	Tuned XGBoost model scores the aligned feature vector and returns a high-risk prediction with probability.	ML Core (best_model.pkl)	Done	High
6	Result Display with Probability	Renders an Approved / Needs Review result along with the underlying risk probability percentage.	Frontend (result.html)	Done	Medium
7	Model Training & Evaluation Pipeline	Full notebook pipeline covering EDA, preprocessing, four-model comparison, and hyperparameter tuning (RandomizedSearchCV).	Notebook (Kaggle / VS Code)	Done	High
8	Error Handling for Unseen Values	Gracefully handles categorical values not seen during training (e.g. rare occupations) without crashing the server.	Backend (app.py)	Done	Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	8
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	8

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Applicant Data Input Form, Result Display with Probability
2	Backend / Logic	5	Categorical Encoding & Remapping, Feature Engineering Recreation, Column Alignment, XGBoost Risk Prediction, Error Handling
3	Database / Storage	1	Model Training & Evaluation Pipeline (produces best_model.pkl, model_columns.pkl, categorical_value_map.pkl)
4	API / Integration	0	None - self-contained Flask app with no external API dependencies
5	Security / Authentication	0	Not implemented in this phase - single-user local application